

Signature-Driven Pollen Classification via Tensor-Based Joint Probability Mass Function (PMF) Estimation

Alla Manina¹, Joseph K. Chege¹, Thomas Hornick^{2,3}, Susanne Dunker^{2,3}, Arie Yeredor⁴, and Martin Haardt¹

¹Communications Research Laboratory, Technische Universität Ilmenau, Ilmenau, Germany

²Helmholtz-Centre for Environmental Research – UFZ, Leipzig, Germany

³German Centre for Integrative Biodiversity Research – iDiv, Halle-Jena-Leipzig, Germany

⁴School of Electrical Engineering, Tel Aviv University, Tel Aviv, Israel

Email: {alla.manina, joseph.chege, martin.haardt}@tu-ilmenau.de, {thomas.hornick, susanne.dunker}@ufz.de, ariey@tauex.tau.ac.il

Abstract—Accurate pollen classification is essential for aerobiology, forecasting of allergenic pollen, and environmental monitoring, yet remains challenging due to high-dimensional feature spaces and overlapping species characteristics. Traditional approaches rely on manual microscopic annotation or fully supervised classifiers, both of which scale poorly with dataset size and labeling effort. This paper proposes a signature-driven framework for pollen classification based on low-rank modeling of joint probability mass functions (PMFs). Each pollen type is represented by a nonnegative CPD of the respective joint PDF of the measured features, estimated from a subset of pollen samples (signatures), without using the labels as supervised training targets for the classification rule. After the signature models are estimated, previously unseen samples are assigned using a signature-driven similarity rule. In addition, we introduce a MAP-log classifier that applies the classical MAP decision rule to the estimated class-conditional joint PMFs in the logarithmic domain. Using a three-class multispectral imaging flow cytometry (MIFC) pollen dataset, we systematically vary the fraction of signature data and evaluate the performance against the naïve Bayes and random forest benchmarks. Since the effectiveness of the scheme depends on the quality of the estimated signatures, we additionally conduct a synthetic study based on the Factor Match Score (FMS), which quantifies how signature quality degrades as the number of available samples to compute the signatures decreases. This provides a practical way to assess, *a priori*, whether enough signature data is available to obtain a reliable classification performance. Our experimental results show that joint-PMF-based signatures form an effective, label-efficient, and explainable alternative for pollen classification, offering a promising alternative to existing benchmark algorithms.

Index Terms—PMF estimation, tensor decomposition, variational Bayesian inference, pollen classification

I. INTRODUCTION

The joint probability mass function (PMF) of a collection of discrete random variables is a key object in many data-analytic tasks, including classification and recommender systems [1]. Direct estimation of such joint PMFs by multidimensional histograms is quickly rendered impractical, as the number of parameters and the required sample size increase exponentially with the number of variables. To circumvent this curse of dimensionality, recent work models the joint PMF as a low-rank tensor and employs a canonical polyadic decomposition (CPD) to obtain a compact representation [1]–[3]. A central challenge in this framework is that the tensor rank is typically unknown in advance. Variational Bayes (VB)-based algorithms have been developed that jointly estimate the PMF and infer the rank from the data, thereby avoiding manual model order selection and enhancing scalability [4]. Such low-rank PMF tensor models with automatic rank detection have already been shown to be effective for analytical flow cytometry data, where they enable the characterization of complex cell populations from high-dimensional measurements [4], [5].

Quantifying pollen concentrations with high reliability is a key requirement in contexts such as plant–pollinator monitoring, paleo-

climate and environmental reconstruction, food quality control, and forecasting of allergy risk [6]. In routine practice, however, pollen counts are still predominantly obtained by light microscopy, a procedure that is time-consuming, depends on specialized expertise, and inherently limits sample throughput [7], [8]. Molecular non-optical techniques, in particular DNA metabarcoding, can process larger numbers of samples but face intrinsic difficulties when translating sequence read numbers into unbiased estimates of relative pollen abundances [9]. In parallel, workflows that combine multispectral imaging flow cytometry (MIFC) with deep learning have shown that automated identification of pollen species and extraction of their traits is feasible at the single-particle level [10], [11].

Recent results indicate that low-rank joint PMF estimation can already be used to quantify pollen proportions in mixtures with a promising performance [12]. Building on this, the present work investigates the further applicability of joint PMF modeling by extending the scenario in [12] to a signature-driven classification scheme. In a first step, joint PMFs are estimated for pure pollen samples and interpreted as characteristic signatures. These signatures are then used in a subsequent step to assign mixture components to their corresponding pollen types, without requiring access to the true labels of the mixed samples. Because this strategy operates directly in the original feature space and does not rely on additional latent representations, it offers a higher degree of interpretability than conventional approaches such as Random Forest-based classifiers, while still exploiting the expressive capabilities of low-rank PMF tensor models.

II. LOW-RANK TENSOR MODELING OF JOINT PMFs

Consider a collection $\mathbf{X} = \{X_1, \dots, X_N\}$ of N discrete random variables, each of which takes values in $[1, I_n]$. The joint PMF $\Pr(\mathbf{X})$ can be represented as an N -dimensional tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times \dots \times I_N}$, where each tensor element represents the joint probability of a certain realization of $\mathbf{X}$. In particular, $\mathcal{X}(i_1, \dots, i_N) = \Pr(X_1 = i_1, \dots, X_N = i_N)$, where $i_n = 1, \dots, I_n, \forall n$. In our context, the random variables $\mathbf{X}$ model the characteristics of various pollen types, obtained from, e.g., flow cytometry measurements. Thus, each pollen sample has N characteristics (or features), each of which is discretized into I_n bins.

The PMF tensor $\mathcal{X}$ can be expressed as a sum of R nonnegative rank-one terms via a (nonnegative) canonical polyadic decomposition (CPD) (e.g., [1])

$$\mathcal{X} = \sum_{r=1}^R \lambda_r \mathbf{A}_1(:, r) \circ \mathbf{A}_2(:, r) \circ \dots \circ \mathbf{A}_N(:, r), \quad (1)$$

where $\mathbf{A}_n \in \mathbb{R}^{I_n \times R}$ denotes a factor matrix with nonnegative elements, and $\mathbf{A}_n(:, r)$ is the r th column of $\mathbf{A}_n$. Furthermore, assuming (without loss of generality) that the columns of the factor

matrices have unit norm, the weights are absorbed into the loading vector $\lambda = [\lambda_1, \dots, \lambda_R]^\top$. Using the CPD notation, one entry of the PMF tensor $\mathcal{X}$ is expressed as

$$\mathcal{X}(i_1, \dots, i_N) = \sum_{r=1}^R \lambda_r \prod_{n=1}^N \mathbf{A}_n(i_n, r). \quad (2)$$

There are two main advantages of modelling a PMF tensor using a CPD. Firstly, the CPD greatly reduces the number of parameters required to describe the joint PMF; only $R(1 + \sum_{n=1}^N I_n)$ rather than $\prod_{n=1}^N I_n$ in the case of the full tensor. Secondly, the CPD model of a PMF tensor admits an insightful statistical interpretation as a latent variable model, specifically the naïve Bayes model [1], [13]. In this model, the random variables $\mathbf{X}$ are independent given a latent variable H with R states. Therefore, the joint probability of a realization of $\mathbf{X}$ is

$$\Pr(X_1 = i_1, \dots, X_N = i_N) = \sum_{r=1}^R \Pr(H = r) \prod_{n=1}^N \Pr(X_n = i_n | H = r). \quad (3)$$

Comparing (2) and (3) reveals that $\lambda_r = \Pr(H = r)$ and $\mathbf{A}_n(i_n, r) = \Pr(X_n = i_n | H = r)$, implying that the CPD components are constrained to the probability simplex, i.e., having nonnegativity ($\lambda > \mathbf{0}$, $\mathbf{A}_n \geq \mathbf{0}$) and sum-to-one ($\mathbf{1}^\top \lambda = 1$, $\mathbf{1}^\top \mathbf{A}_n = \mathbf{1}^\top$) constraints, where $\mathbf{1}$ is an all-ones vector.

Several approaches for estimating the CPD components have been proposed in the literature, ranging from coupled factorization of lower-order marginal tensors [1], [14] to maximum-likelihood [2], [3] and Bayesian methods [4]. In this work, we adopt the variational Bayes PMF estimation approach (VB-PMF) presented in [4], which jointly estimates the CPD components and the rank from the observed data using Bayesian inference. Note that for real-world data, the rank of the joint PMF tensor is not known. Usually, the rank is treated as a parameter to be tuned and is selected via cross-validation, which could be computationally expensive. In contrast, the VB-PMF algorithm infers the appropriate rank from the data without cross-validation. Due to these advantages, we use the VB-PMF algorithm to estimate the joint PMF. However, we emphasize that our proposed classification approaches are independent of the PMF estimation algorithm; any method capable of estimating a nonnegative CPD consistent with (1) could be used instead of VB-PMF.

III. SIGNATURE-DRIVEN POLLEN CLASSIFICATION

Building on the low-rank tensor modeling of joint PMFs in (1)-(2) and their naïve Bayes interpretation, we introduce a signature-driven classification scheme for multiple pollen types. Consider C pollen types, indexed by $c \in \{1, \dots, C\}$, each characterized by the same collection of N discrete random variables $\mathbf{X} = \{X_1, \dots, X_N\}$, where $X_n \in [1, I_n]$ denotes the bin index of the n th feature. For each pollen type c , the joint PMF of $\mathbf{X}$ is represented by an N -way PMF tensor $\mathcal{X}^{(c)} \in \mathbb{R}^{I_1 \times \dots \times I_N}$, which admits a nonnegative CPD of the form

$$\mathcal{X}^{(c)}(i_1, \dots, i_N) = \sum_{r=1}^{R_c} \lambda_{c,r} \prod_{n=1}^N \mathbf{A}_{c,n}(i_n, r), \quad i_n = 1, \dots, I_n, \quad (4)$$

with factor matrices $\mathbf{A}_{c,n} \in \mathbb{R}^{I_n \times R_c}$ and loading vector $\lambda_c = [\lambda_{c,1}, \dots, \lambda_{c,R_c}]^\top \in \mathbb{R}^{R_c}$. The dominant components in (2), i.e., the columns of $\mathbf{A}_{c,n}$, $n = 1, \dots, N$, are interpreted as conditional PMFs,

$$\mathbf{A}_{c,n}(:, r) = \Pr(X_n | H_c = r), \quad r = 1, \dots, R_c, \quad (5)$$

where H_c is a latent variable associated with pollen type c , and

$$\lambda_{c,r} = \Pr(H_c = r), \quad r = 1, \dots, R_c. \quad (6)$$

For each pollen type c , the available discretized samples are partitioned into two disjoint subsets: a *signature set*, used to estimate the parameters $\{\mathbf{A}_{c,1}, \dots, \mathbf{A}_{c,N}, \lambda_c\}$, referred to as the *signature model* of type c , of (4), and a *mixture set*, reserved exclusively for classification.

After estimating the joint PMF of each pollen type individually from the signature set, we obtain the low-rank components

$$\{\hat{\mathbf{A}}_{c,1}, \dots, \hat{\mathbf{A}}_{c,N}, \hat{\lambda}_c\}_{c=1}^C. \quad (7)$$

Next, all mixture samples from the C pollen types are pooled into a single dataset. Each mixture sample j is represented by an N -dimensional integer vector

$$\mathbf{k}^{(j)} = (k_1^{(j)}, \dots, k_N^{(j)})^\top, \quad k_n^{(j)} \in \{1, \dots, I_n\}, \quad (8)$$

collecting the bin indices of the N features for that sample.

The proposed classification rule compares each observed feature bin directly with the corresponding entries of the signature model. For a given pollen type c , the contribution of feature n to the similarity between sample j and type c is obtained from the entries $\mathbf{A}_{c,n}(k_n^{(j)}, r)$ of the factor matrix $\mathbf{A}_{c,n}$, weighted by the corresponding $\lambda_{c,r}$. The resulting score for pollen type c and mixture sample j is defined as

$$d_c^{(j)} = \frac{1}{N} \sum_{n=1}^N \sum_{r=1}^{R_c} \left(\frac{\mathbf{A}_{c,n}(k_n^{(j)}, r)}{\|\mathbf{A}_{c,n}(:, r)\|_2} \lambda_{c,r} \right), \quad c = 1, \dots, C, \quad (9)$$

where $\|\cdot\|_2$ denotes the Euclidean norm. Collecting all scores in a matrix yields

$$\mathbf{D} = [\mathbf{d}^{(1)}, \dots, \mathbf{d}^{(T)}] \in \mathbb{R}^{C \times T}, \quad (10)$$

where $\mathbf{d}^{(j)} = [d_1^{(j)}, \dots, d_C^{(j)}]^\top \in \mathbb{R}^C$ denotes the vector of class scores for mixture sample j from (9), $j = 1, \dots, T$.

The predicted class index for mixture sample j is obtained via

$$\hat{c}^{(j)} = \arg \max_{c \in \{1, \dots, C\}} d_c^{(j)}. \quad (11)$$

If a label set $\mathcal{L} = \{\textit{Betula}, \textit{Carpinus}, \dots\}$ is specified, the index $\hat{c}^{(j)}$ is mapped to the corresponding pollen type name.

A natural alternative to (9) would be a Maximum A Posteriori (MAP)-type decision rule based directly on the class-conditional joint PMF,

$$p_c(\mathbf{k}^{(j)}) = \sum_{r=1}^{R_c} \lambda_{c,r} \prod_{n=1}^N \mathbf{A}_{c,n}(k_n^{(j)}, r). \quad (12)$$

However, when the number of features is high, evaluating this likelihood requires multiplying several hundred probability values. This is numerically delicate, since the resulting products can become extremely small and may underflow in finite-precision arithmetic. Although this issue can be alleviated by evaluating the likelihood in the logarithmic domain, such a formulation requires additional treatment of zero or near-zero factor entries. The impact of this effect and the corresponding benefit of the log-domain formulation are shown in Section IV.

At the same time, the effectiveness of the proposed schemes therefore depend on the quality of the estimated signatures, which in turn is governed by the amount of data available for each pollen type. When the signature sets are very small, it becomes more challenging for the estimated joint PMF models to capture the underlying variability, although the resulting classifier can still achieve a robust performance

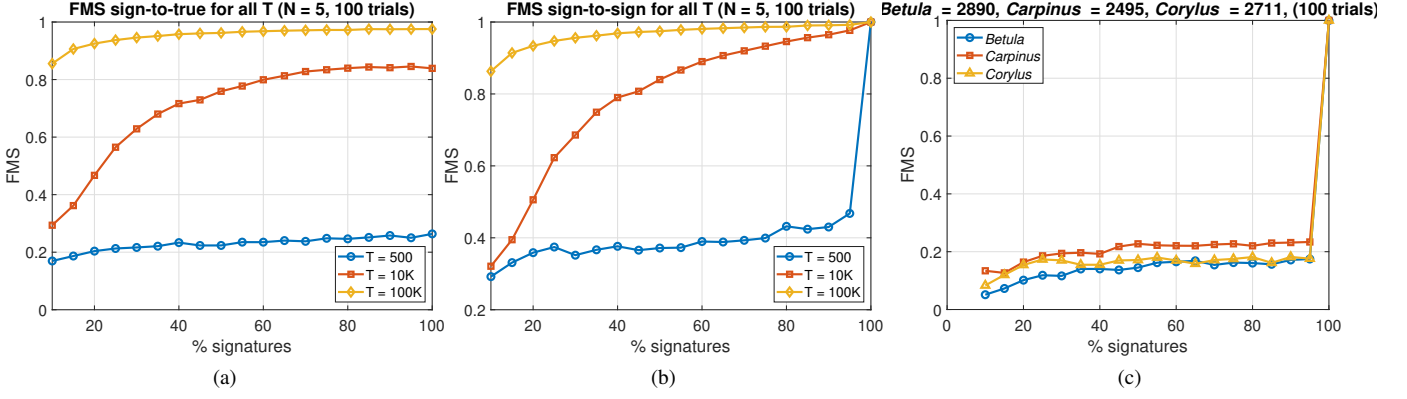


Fig. 1: FMS versus percentage of signatures in the synthetic experiments ($N = 5$ features) and the experiment in the real MIFC dataset ($N = 543$).

in practice. This naturally raises the question of how to assess, in advance, whether a given amount of signature data is sufficient for reliable pollen classification. In the following section, we address this question by studying how the signature size affects the accuracy of the estimated joint PMFs with simulation-based experiments on synthetic data.

IV. RESULTS AND DISCUSSIONS

To evaluate the performance of the signature-driven classification schemes, two experiments are conducted. Using the synthetic dataset, the first experiment was conducted to analyze how the quality of the estimated joint PMFs increases as the fraction of signature data is increased. Then, the classification performance was assessed on a real MIFC pollen dataset.

The dataset consists of measurements of three pollen types with 577 extracted features obtained from MIFC analyses [10], [11]. The features summarize estimates of size, shape, signal strength, texture, and spatial location. The dataset contains the following pollen types: *Betula* with $T = 2890$ samples, *Carpinus* with $T = 2495$ samples, and *Corylus* with $T = 2711$ samples. Before further processing, low-variance features were removed across all pollen types, resulting in $N = 543$ retained features. These features were then discretized into $I_n = 20$ bins.

To assess how well the joint PMF factors can be recovered from limited signature data, we evaluate the *Factor Match Score* (FMS) [15], which measures the similarity between two CPD models up to permutation of their components. Given a true model with weights λ_r and factors $\{\mathbf{a}_r^{(n)}\}_{n=1}^N$, and an estimated model with weights $\hat{\lambda}_r$ and factors $\{\hat{\mathbf{a}}_r^{(n)}\}_{n=1}^N$, the FMS is defined as

$$\text{FMS} = \frac{1}{R} \sum_{r=1}^R \left(1 - \frac{|\xi_r - \hat{\xi}_r|}{\max\{\xi_r, \hat{\xi}_r\}} \right) \prod_{n=1}^N \frac{\mathbf{a}_r^{(n)\top} \hat{\mathbf{a}}_r^{(n)}}{\|\hat{\mathbf{a}}_r^{(n)}\| \|\hat{\mathbf{a}}_r^{(n)}\|}, \quad (13)$$

where

$$\xi_r = \lambda_r \prod_{n=1}^N \|\mathbf{a}_r^{(n)}\|, \quad \hat{\xi}_r = \hat{\lambda}_r \prod_{n=1}^N \|\hat{\mathbf{a}}_r^{(n)}\|. \quad (14)$$

The FMS takes values in $[0, 1]$, with 1 indicating a perfect match between the two decompositions (up to permutation).

In a synthetic-data experiment, we generate a ground-truth non-negative CPD with $N = 5$ features, $I_n = 20$ bins per feature, and rank $R = 5$. The true factor matrices $\{\mathbf{A}_{\text{true},n}\}_{n=1}^N$ are drawn from $U(0, 1)$ and normalized to fulfill the simplex constraints, whereas the loading vector is fixed as $\lambda_{\text{true}} = [0.3, 0.1, 0.2, 0.15, 0.25]^\top$.

From this joint PMF model we draw independent samples for three total sample sizes, $T \in \{500, 10^4, 10^5\}$. For each T and for each signature proportion in the range from 10% to 100%, a subset of size $T_{\text{sign}} = T \cdot (\text{signature fraction})$ is used as a synthetic signature set, and a joint PMF is estimated via VB-PMF [4] with initial rank $R = 50$. For every proportion and each Monte Carlo trial (100 trials for each $T \in \{500, 10^4, 10^5\}$), we compute (i) the FMS between the estimated model and the ground-truth model (“sign-to-true” in 1a) and (ii) the FMS between the estimated model and a reference model obtained from the full 100% signature set (“sign-to-sign” in 1b) that can be obtained in practice when the ground-truth model is not known.

In Figs. 1a–1b, it can be seen that, for $T = 10^5$, the FMS remains high even for small signature fractions, indicating good quality of the signature estimates. A similar tendency is visible for $T = 10^4$ samples, where the FMS stays at a reasonably high level across the range of the low signature proportions. In contrast, with $T = 500$ samples in total, the FMS drops significantly already at 95% signatures, suggesting that the available data are insufficient to stably estimate the joint PMFs. A similar behavior is observed for the Kullback–Leibler divergence (KLD) between the tensor reconstructed from a reduced signature fraction and that reconstructed from the full 100% signature set. Observing such a pattern in practice would therefore indicate that the overall amount of data is not enough for a reliable classification.

Fig. 1c shows the sign-to-sign FMS for the real MIFC dataset as a function of the signature fraction for the three pollen types. Comparing Fig. 1c with Figs. 1a–1b, it can be observed that the real MIFC data behave similarly to the low-sample synthetic cases. This indicates that the current pure-sample MIFC dataset has not yet reached the regime in which the joint PMF signatures can be estimated with the same accuracy as in the large-sample synthetic case ($T = 10^5$), although, as will be shown further, the estimated signatures are already sufficient to yield a robust classification performance.

For the real-data classification experiment, we consider a fixed signature proportion of 95% per pollen type. For each pollen type, the corresponding discretized samples are randomly split into two disjoint subsets: a *signatures* set, containing 95% of the samples, and a *mixture* set, containing the remaining 5%. The signature set is used to estimate the joint PMF model for each pollen type and to train the benchmark classifiers, whereas the mixture set is reserved exclusively for testing. This random split is repeated over 100 independent Monte Carlo trials.

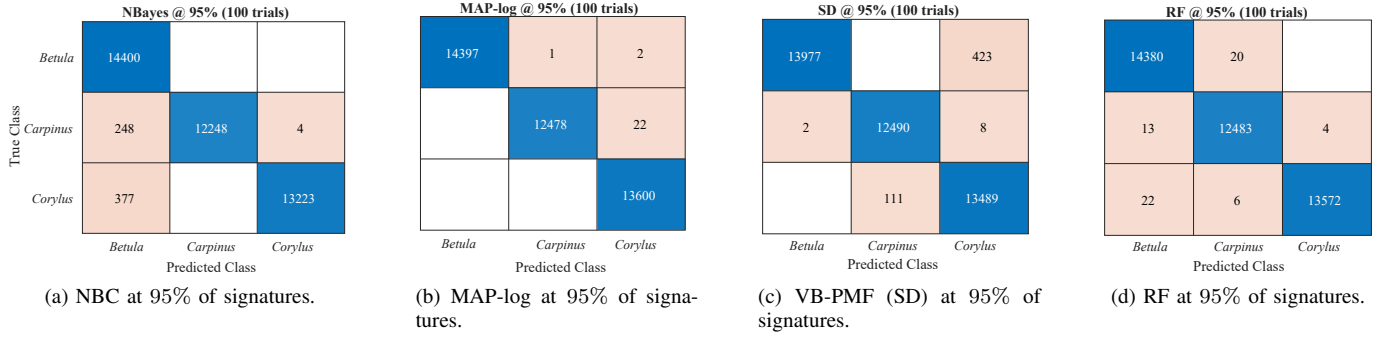


Fig. 2: Summed confusion matrices over 100 trials for NBC, MAP-log, RF, and the signature-driven classifier VB-PMF (SD) at a 95% signature proportion.

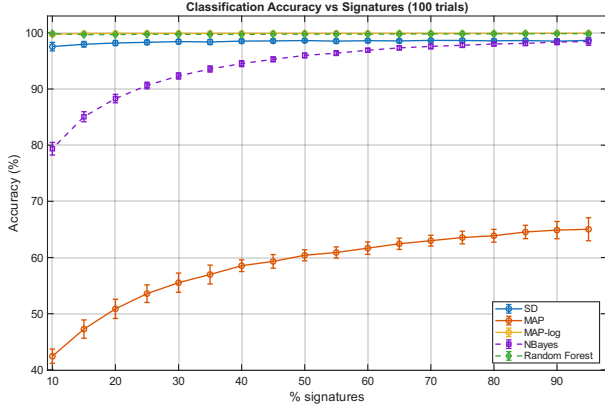


Fig. 3: Classification accuracy vs signatures % used for training.

As benchmark classification algorithms on the MIFC pollen dataset, we use the naïve Bayes classifier (NBC) and the random forest (RF) algorithm. All benchmark classifiers are trained on the same signature data (fixed at 95% of the available samples per pollen type) and evaluated on the same mixture data with known labels. The RF employs an ensemble of 300 decision trees and treats all features as categorical. The signature-driven approaches considered here are VB-PMF (SD), the MAP classifier based on (12), and its log-domain implementation (MAP-log). These methods estimate the joint PMFs of the individual pollen signature datasets with an initial CPD rank of $R = 50$, leading to automatically inferred ranks R_c , $c \in \{1, \dots, C\}$, for each pollen type. In contrast to the benchmark classifiers, the signature-driven methods do not use sample labels as supervised training targets; the labels are used only to group the signature data by pollen type before estimating the corresponding joint PMFs.

The classification performance of the signature-driven and the benchmark algorithms is summarized in confusion matrices. In Fig. 2, the true and predicted labels aggregated over 100 trials are shown for the 95% signature proportion. It can be observed that all methods achieve high accuracy at a 95% signature proportion, with strongly diagonal-dominant confusion matrices. The naïve Bayes classifier (NBC) reaches a per-class recall above 97.2% and a per-class precision close to or above 95.8%. The MAP-log classifier and RF show a nearly perfect performance, with per-class recall and precision close to 100% for all three pollen types. The VB-PMF (SD) classifier also shows a high per-class recall above 97.1% and a per-class precision above 96.9%, respectively. While the benchmark classifiers

exploit fully supervised training on the labeled signature set, VB-PMF (SD) and MAP-log use the signature data only to estimate class-conditional joint PMFs and then apply a fixed probabilistic similarity rules. Thus, no label information is used when learning the signature models themselves; the labels are required only for evaluating the predictions on the mixture samples. In this setting, the macro-averaged F1 score is approximately 98.48% for NBC, 99.94% for MAP-log, 98.68% for VB-PMF (SD), and 99.84% for RF, demonstrating that the signature-driven classifier remains competitive with fully supervised benchmarks.

In Fig. 3, the classification accuracy versus the percentage of signatures used for the training is depicted. Here, we also show the sensitivity of the direct MAP classifier when the likelihood is evaluated in the original probability domain. The direct MAP classifier is strongly affected by the multiplication of many small probability values and yields a substantially lower accuracy. The MAP-log classifier successfully overcomes this challenge showing a comparable performance to the RF benchmark and the signature-driven method (SD) confirming the numerical robustness of the classification rule.

V. CONCLUSIONS

In this paper, we propose signature-driven classification schemes for MIFC pollen data, where each pollen type is modeled by a low-rank joint PMF and the factor matrices act as probabilistic signatures. The FMS study shows that the quality of these signatures improves as more signature data is used, allowing the required signature size for a target accuracy to be estimated in advance. This evaluation strategy can be integrated into the PMF-based pollen quantification framework of [12]. On a real three-class MIFC dataset, the VB-PMF (SD) and MAP-log methods achieve a high classification performance that is competitive with naïve Bayes and random forest classifiers, indicating that joint-PMF-based signatures form an effective alternative for pollen classification. Moreover, with the signature-driven methods, the results are explainable since the most relevant features for these classification decisions can easily be identified.

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REFERENCES

- [1] N. Kargas, N. Sidiropoulos, and X. Fu, “Tensors, Learning, and “Kolmogorov Extension” for Finite-Alphabet Random Vectors,” *IEEE Transactions on Signal Processing*, vol. 66, 2017.
- [2] A. Yeredor and M. Haardt, “Maximum Likelihood Estimation of a Low-Rank Probability Mass Tensor From Partial Observations,” *IEEE Signal Processing Letters*, vol. 26, no. 10, pp. 1551–1555, 2019.
- [3] J. K. Chege, M. J. Grasis, A. Manina, A. Yeredor, and M. Haardt, “Efficient Probability Mass Function Estimation from Partially Observed Data,” in *Proc. 56th Asilomar Conference on Signals, Systems, and Computers*. IEEE, 2022.
- [4] J. K. Chege, M. J. Grasis, A. Yeredor, and M. Haardt, “Bayesian Estimation of a Probability Mass Function Tensor with Automatic Rank Detection,” in *Proc. 9th IEEE International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, 2023.
- [5] P. Flores, J. K. Chege, K. Usevich, M. Haardt, A. Yeredor, and D. Brie, “Probability Mass Function Estimation Approaches with Application to Flow Cytometry Data Analysis,” in *Proc. IEEE 9th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, 2023, pp. 451–455.
- [6] T. Hornick, A. Richter, W. S. Harpole, M. Bastl, S. Bohlmann, A. Bonn, J. Bumberger, P. Dietrich, B. Gemeinholzer, R. Grote *et al.*, “An integrative environmental pollen diversity assessment and its importance for the Sustainable Development Goals,” *Plants, People, Planet*, vol. 4, no. 2, pp. 110–121, 2022.
- [7] G. Erdtman, *Pollen Morphology and Plant Taxonomy*. Brill Archive, 1986.
- [8] K. Holt and K. D. Bennett, “Principles and methods for automated palynology,” *New Phytologist*, vol. 203, no. 3, pp. 735–742, 2014.
- [9] K. Bell, N. de Vere, A. Keller, R. Richardson, A. Gous, K. Burgess, and B. Brosi, “Pollen DNA barcoding current applications and future prospects,” *Genome*, vol. 59, no. 9, pp. 629–640, 2016.
- [10] S. Dunker, E. Motivans, D. Rakosy, D. Boho, P. Mäder, T. Hornick, and T. M. Knight, “Pollen analysis using multispectral imaging flow cytometry and deep learning,” *New Phytologist*, vol. 229, no. 1, pp. 593–606, 2021. [Online]. Available: <https://nph.onlinelibrary.wiley.com/doi/abs/10.1111/nph.16882>
- [11] T. Hornick, W. S. Harpole, and S. Dunker, “High-throughput assessment of anemophilous pollen size and variability using imaging cytometry,” *New Phytologist*, 2025.
- [12] A. Manina, J. K. Chege, T. Hornick, S. Dunker, A. Yeredor, and M. Haardt, “Pollen Quantification Using Bayesian Probability Mass Function Estimation with Automatic Rank Detection,” in *Proc. 59th Asilomar Conference on Signals, Systems, and Computers*. IEEE, 2025.
- [13] M. Ishteva, “Tensors and latent variable models,” in *Latent Variable Analysis and Signal Separation*. Springer, 2015, pp. 49–55.
- [14] A. Yeredor and M. Haardt, “Estimation of a Low-Rank Probability-Tensor from Sample Sub-Tensors via Joint Factorization Minimizing the Kullback-Leibler Divergence,” in *Proc. 27th IEEE European Signal Processing Conference (EUSIPCO)*, A Coruna, Spain, Sep. 2019.
- [15] E. C. Chi and T. G. Kolda, “On Tensors, Sparsity, and Nonnegative Factorizations,” *SIAM J. Matrix Anal. & Appl.*, vol. 33, no. 4, pp. 1272–1299, Jan. 2012.